

Department: Computer Science
Specialization: Software Engineering
Generation: 11, Group: 01

Backend Project Proposal

Course: Backend Development
Lecturer: Mr. Ngin Kimlong

Project: Mindful Study

Concept: Smart Planner & Burnout Prevention System

Team 4

Team Name: Tri-K

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I. Introduction

- **Background**

Mindful Study, a **Study Planner & Burnout Prevention System** is a web-based application designed to help university students manage their academic responsibilities while maintaining a healthy study-life balance. The system combines task management, study scheduling, and burnout monitoring into a single platform.

Mindful Study is created because students often face difficulties organizing assignments, tracking deadlines, and balancing academic workloads. These challenges can lead to stress, decreased productivity, and burnout.

- **Objectives**

- **General Objective**

Develop a web-based Study Planner & Burnout Prevention System that helps students organize their academic workload and reduce burnout risk.

- **Specific Objective for Backend Development**

- Implement secure user authentication and authorization.
- Provide task and deadline management features.
- Allow students to schedule study sessions.
- Monitor stress and burnout indicators.
- Generate dashboard insights regarding academic progress.

- **Impact**

The system is expected to:

- Improve student time management.
- Reduce missed deadlines.
- Encourage healthier study habits.
- Increase awareness of burnout symptoms.
- Support academic success through structured planning.

II. Methodology & Tools

- **Development Methodology**

- Agile Scrum**

- The project will follow Agile Scrum practices with iterative development, regular progress reviews, and continuous testing.

- **Software Architecture**

- **MVC Architecture**

- The backend will follow the Model-View-Controller (MVC) architecture to ensure maintainability and separation of concerns.

- **REST API**

- The application will expose RESTful API endpoints developed using Node.js and Express.js.

- **Database Design**

- **Database**

- MySQL will be used as the primary relational database management system.

- **Normalization**

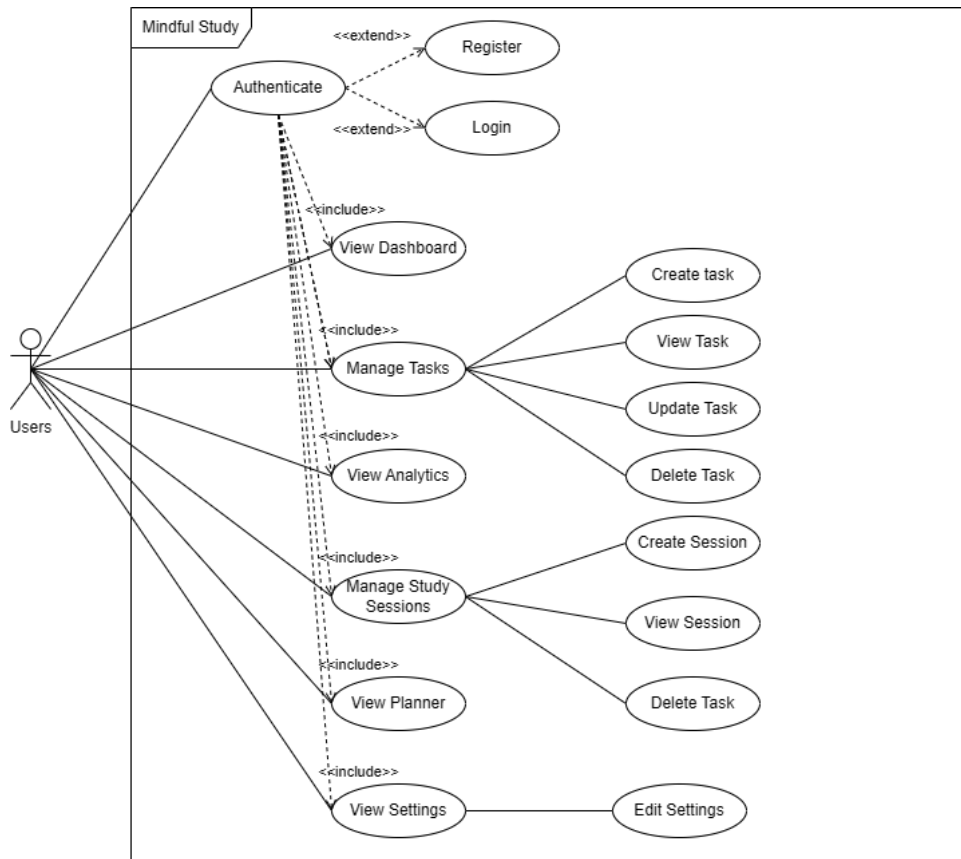
- Database tables are normalized up to Third Normal Form (3NF):

- Entities and Attributes:**

- **Users:** user_id, full_name, email, password, role, created_at
 - **Categories:** category_id, name, color
 - **Tasks:** task_id, user_id, category_id, title, description, priority, status, due_date, created_at, completed_at
 - **StudySessions:** session_id, user_id, task_id, start_time, end_time, duration_minutes (Since duration_minutes is a derived attribute from which $\text{duration_minutes} = \text{end_time} - \text{start_time}$, it is stored as a denormalized attribute for reporting and dashboard calculations)
 - **Notifications:** notification, user_id, message, type, is_read, created_at

- **Recommendations:** recommendation, user_id, message, recommendation type, created_at
- **MoodLogs:** mood_id, user_id, mood_level, note, created_at
- **UserSettings:** setting_id, user_id, notification_enabled, dark_mode

• UML Design



• Development Tools

- **Frontend**
React JS
- **Backend**
Node.js, Express.js
- **Database**
Aiven MySQL
- **Authentication**
Jason Web Token (JWT)

- **API Testing**
Postman
- **API Documentation**
Swagger UI

IV. Plan

- **Timeline**
 - **Week 2 – Week 4**
 - Project Ideation
 - UX research conduct (Interview, Survey)
 - Concept-refining and defining indirect compositions (Target users, Pain points, Persona ...)
 - Research existing study planner systems
 - **Week 5**
 - Requirement Analysis
 - User Storyboard
 - Use Case Diagram
 - **Week 6**
 - Figma Prototype Design
 - Project Proposal
 - Database design
 - Activity Diagram
 - **Week 7**
 - Frontend Implementation
 - Authentication module implementation
 - User management APIs
 - Routing and Middleware
 - Class Diagram
 - **Week 8**
 - Frontend Implementation
 - Task management APIs

- Study session APIs
 - Mood assessment APIs
- **Week 9**
 - Dashboard APIs
 - Frontend Integration
 - Database Queries
 - Database roles and users' privilege
- **Week 10**
 - Swagger documentation
 - Testing and bug fixing
 - Final improvements
- **Week 11**
 - Final Presentation
- **WBS (Work Breakdown Structure)**
 - 1. Project Planning**
 - 1.1 Project ideation
 - 1.2 Requirement gathering
 - 1.3 User research
 - 1.4 Proposal preparation
 - 2. System Design**
 - 2.1 User story creation
 - 2.2 Use Case Diagram
 - 2.3 Activity Diagram
 - 2.4 Class Diagram
 - 2.5 Database design
 - 2.6 Database normalization (3NF)
 - 2.7 Figma prototype design
 - 3. Backend Development**
 - 3.1 Authentication module
 - User registration
 - User login
 - JWT authentication

3.2 Task management module

- Create task
- View tasks
- Update task
- Delete task

3.3 Study session module

- Create study session
- View study sessions
- Update study sessions
- Delete study sessions

3.4 Mood assessment module

- Log mood
- View mood history

3.5 Dashboard module

- Statistics API
- Progress tracking API

3.6 Middleware

- Authentication middleware
- Error handling middleware

4. Database Implementation

4.1 Create database schema

4.2 Create tables and relationships

4.3 Implement constraints

4.4 Create user roles and privileges

4.5 Develop SQL queries

5. Frontend Development

5.1 Authentication pages

5.2 Dashboard page

5.3 Task management page

5.4 Study session page

5.5 Mood tracking page

5.6 Settings page

6. Testing & Documentation

6.1 API testing using Postman

6.2 Swagger documentation

6.3 Integration testing

6.4 Bug fixing

6.5 Final documentation

7. Deployment & Presentation

7.1 Final system integration

7.2 Demo preparation

7.3 Presentation preparation

7.4 Project defense

V. Responsibilities

- **Kong Sothearith**

- Figma Design
- React JS Design
- Database Design
- Class Diagram
- Authentication module
- User APIs
- JWT implementation
- API Testing

- **Kong Theara**

- Figma Design
- Use Case Diagram
- Task management APIs
- Study session APIs
- API testing
- Database Queries (DML, DQL)

- **Kov Cheaching**

- React JS Design
- Activity Diagram
- Mood assessment APIs

- Swagger documentation
- Middleware with error-handling
- Database role and users' privileges

VII. Expected Result

The completed system will provide:

- o Secure authentication and authorization
- o Task management through CRUD operations
- o Study session scheduling
- o Burnout monitoring and assessment
- o Dashboard reporting and progress tracking
- o Fully documented REST APIs
- o Integration between ReactJS frontend and Node.js backend.

VIII. Demo Prototype

Figma Link:

https://www.figma.com/proto/d9Hj1BjK5avufAlr7PbXzE/Smart_Planner-BPS?node-id=0-1&t=FQYcBttG3sbfk9Bv-1